

IT Disaster Recovery Planning

[Source: Page 415 to 423, CISSP for Dummies]

A variety of disasters can beset an organization's business operations. They fall into two main categories: natural and man-made. Like,

Fires and explosions, Earthquakes, Storms, Floods, Pandemics, Accidents, Crime, War and terrorism, Cyber-attacks, Civil disturbances and such others.

Disasters can affect businesses in a lot of ways:

- Damage to business buildings.
- Damage to business records.
- Damage to business equipment.
- Damage to communications.
- Damage to public utilities.
- Damage to transportation systems.
- Injuries and loss of life.
- Indirect damage: suppliers and customers.

Disaster Recovery Plan (DRP)

Response

Emergency response teams must be prepared for every reasonably possible scenario. Organizations must document all the types of responses so that the response teams know what to do. The emergency response documentation consists of two major parts: how to respond to each type of incident, and the most up-to-date facts about the facilities and equipment that the organization uses.

Salvage

The salvage team is concerned with restoring full functionality to the damaged facility. The salvage team is primarily concerned with the restoration of a facility and its return to operational readiness.

Recovery

Recovery comprises equipping the Business Continuity Plan team with any logistics, supplies, or coordination in order to get alternate functional sites up and running. This activity should be heavily scripted, with lots of procedures and checklists in order to ensure that every detail is handled.

Financial readiness

The salvage and recovery operations can cost a lot of money. The organization must prepare for potentially large expenses (at least several times the normal monthly operating cost) to restore operations to the original facility. Financial readiness can take several forms, including: Insurance, Cash reserves, Line of credit, Pre-purchased assets, Letter of Agreements, Standby assets etc.

Personnel

People are the most important resource in any organization. As such, disaster response must place human life above all other considerations when developing disaster response plans and when emergency responders are taking action after a disaster strikes.

Communications

A critical component is the communications plan. Employees need to be notified about closed facilities and any special work instructions (such as an alternate location to report for work).

Assessment

When a disaster strikes, an organization's DRP team needs to include procedures to assess damage to buildings and equipment. First, the response team needs to examine buildings and equipment, to determine which assets are total losses, which are repairable, and which are still usable.

Restoration

The ultimate objective of the disaster recovery team is the restoration of work facilities with their required assets, so that business may return to normal. Prior to full restoration, business operations may be conducted in temporary facilities, possibly by alternate personnel who may be other employees or contractors hired to fill in and help out. These temporary facilities may be located either near the original facilities or some distance away.

Training and awareness

An organization's ability to effectively respond to a disaster is highly dependent on its advance preparations. In addition to the development of high quality, workable disaster recovery and business continuity plans that are kept up to date, the next most important part is making sure that employees and other needed personnel are periodically trained in the actual response and continuity procedures.

Test Disaster Recovery Plans

By the time that an organization has created a Disaster Recovery Plan (DRP), they'd test the DRP to make sure that it really works when an actual disaster strikes.

The following outline DRP testing methods:

Read-through

A read-through (or checklist) test is a detailed review of DRP documents, performed by individuals on their own. The purpose of a read-through test is to identify inaccuracies, errors, and omissions in DRP documentation.

Walkthrough or tabletop

A walkthrough (or tabletop or structured walkthrough) test is a team approach to the read-through test. Here, several business and technology experts in the organization gather to "walk" through the DRP. A moderator or facilitator leads participants to discuss each step in the DRP so that they can identify issues and opportunities for making the DRP more accurate and complete. Group discussions usually help to identify issues that people will not find when working on their own.

Simulation

In a simulation test, all the designated disaster recovery personnel practice going through the motions associated with a real recovery.

Parallel

A parallel test involves performing all the steps of a real recovery, except that you keep the real, live production systems running. The actual production systems run in parallel with the disaster recovery systems.

Full interruption (or cutover)

A full interruption (or cutover) test is similar to a parallel test except that in a full interruption test, a function's primary systems are actually shut off or disconnected.